

Akhil Mohammad

Integrated B.E. in Computer Science and M.Sc. in Economics, 2027

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Education

Birla Institute of Technology and Science, Pilani – Hyderabad Campus

2022 – 2027

CGPA: 7.71/10, Integrated B.E. in Computer Science and M.Sc. in Economics

Experience

Defence Research and Development Organisation (DRDO), Hyderabad

May 2025 – Jul 2025

- Built a **SAR-based ship detection** system integrating deep learning inference with AIS metadata for identity-aware maritime surveillance, with interactive analysis and visualization via a **Streamlit GUI**.
- Engineered custom preprocessing, annotation reconstruction, and validation pipelines to convert heterogeneous SAR datasets, including OpenSAR and SSDD, into YOLO-ready detection data; trained and evaluated YOLO-based models, achieving **strong baseline localization performance on SSDD (mAP@0.5 = 0.875)**.

National Centre for Polar and Ocean Research, Goa

May 2024 – Jul 2024

- Developed and evaluated time-series forecasting models (**CNNs, RNNs, GRUs, and LSTMs**) on univariate weather data with **TensorFlow and Keras**.
- **Designed and fine-tuned 1D-CNN models** to extract temporal features from sequential weather observations collected at Bharati Station, Antarctica.
- Implemented a **Temporal Convolutional Network** and achieved a Normalized RMSE score of 0.074.

Anukta Infotech Pvt. Ltd.

May 2025 – Jun 2025

- Designed a **feedback-driven RAG system** that updates its knowledge base using user feedback by re-embedding corrected information into a **FAISS vector index**, enabling iterative improvement of responses.
- Implemented an end-to-end RAG pipeline including document chunking, semantic retrieval, LLM-based response generation, and feedback-conditioned knowledge updates using local LLM inference.
- Built a **natural-language-to-SQL system** that converts informal user queries into executable SQL, runs them on structured databases, and presents results as **Pandas dataframes in Streamlit**.

AvidInvent, Independent Web Developer

Jul 2025 – Oct 2025

- Restored the original WordPress/PHP website from downtime.
- Built a new website from scratch using **Astro** and deployed it on **Netlify**.

Projects

Distributed KV Store with Tunable Consistency – Distributed Systems

Mar 2026

- Designed and implemented a distributed key-value store with **leader election** and asynchronous replication across multiple nodes in Python
- Developed dual-mode read semantics (eventual vs consistency) using **multi-node querying** and **version-based reconciliation** to simulate quorum reads.
- Added failure detection and automatic leader re-election using **heartbeat monitoring**

Hindi Information Retrieval Engine – Search Systems

Sep 2024 – Nov 2024

- Engineered an end-to-end IR system with a positional **inverted index** over a 600K+ token Hindi corpus, enabling efficient Boolean and phrase query processing
- Optimized storage and retrieval using **gap encoding + variable-byte compression**, achieving $3.5\times$ index size reduction and improving memory efficiency for large-scale indexing
- Implemented tolerant retrieval with **k-gram indexing** and rule-based transliteration, supporting wildcard and cross-script queries

Skills

- **Backend:** REST APIs, Flask, SQL, Postgres
- **Distributed Systems & Databases:** Replication, Consistency Models, Inverted Index, Query Processing
- **Cloud & Infra:** Kubernetes (basic), AWS (EC2, S3 – basic)
- **Machine Learning:** Time-series, Computer Vision, TensorFlow, PyTorch, Scikit-learn
- **Programming Languages:** Python, C++, Java, C, R, JavaScript